



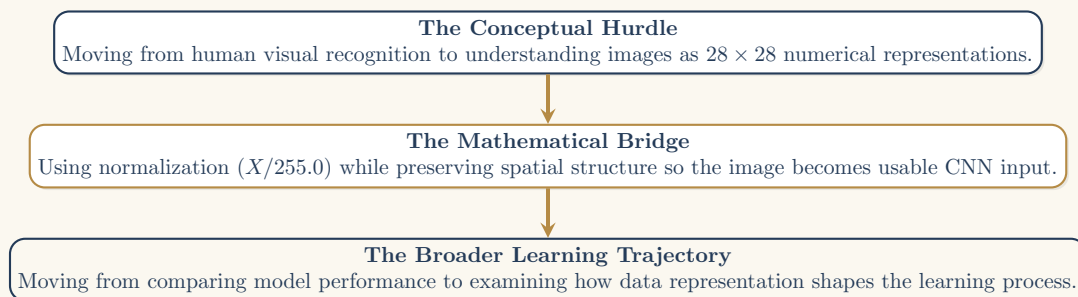
Laboratory Reflection

MNIST Data Preparation and the CNN Workflow

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Before beginning this laboratory, I understood the Convolutional Neural Network (CNN) workflow mainly in terms of training and evaluating a model. I knew that the MNIST dataset contained images of handwritten digits, but I had not fully considered how the mathematical representation of those images affects every later stage. I had previously worked on a Hotel Booking Demand classification project in another course. That project involved preparing a dataset, dividing the data for training and evaluation, training multiple classification models, comparing evaluation metrics, and examining ensemble approaches. That experience gave me a foundation for understanding machine learning as a connected workflow, but this laboratory made me examine an earlier part of that workflow more closely: how the original data is represented before the model can learn from it.

My primary contribution to this laboratory was the MNIST data pipeline. I worked with the 28×28 grayscale image structure, normalized pixel values, and preserved the spatial arrangement of the images. Dividing the original pixel intensities by 255.0 converted the values from the 0–255 range to a standardized 0.0–1.0 range. This made preprocessing tangible to me. The visual handwriting did not change, but its numerical representation became more suitable for the mathematical operations performed during training. I began to understand that preparing data is not simply getting a dataset into the correct format. It establishes the representation that the model receives and processes.



My main challenge during this laboratory was cognitive: shifting from seeing images as a person does to understanding them as multidimensional numerical tensors. I can recognize a handwritten digit at a glance, but a model receives an array of pixel intensities. Recognizing this difference helped me understand why normalization, reshaping, and label preparation are important. Preprocessing became a bridge between the original image and the mathematical operations of the network rather than an administrative step that simply occurs before training. I also came to understand why spatial structure matters in a CNN. A handwritten digit is not merely a flat list of unrelated pixel values. The positions of those values create edges, curves, and other visual patterns. Keeping the images in their 28×28 arrangement allows convolutional layers to examine localized regions and learn from those relationships. This helped me connect the preprocessing stage directly to the later Conv2D operations, rather than treating each section of the notebook as an isolated coding task. The representation created during preparation becomes

the starting point for the feature-learning process.

Revisiting my earlier Hotel Booking Demand work helped me recognize this distinction more clearly. In that project, completed for another course, the emphasis was on a classification problem in which different models could be trained and compared. The workflow included data preparation, training and validation, evaluation with classification metrics, and comparison of approaches such as Logistic Regression, Decision Tree, Random Forest, XGBoost, KNN, and SVC. Ensemble approaches were also examined. The central analytical concern was therefore strongly connected to model performance: how well did different approaches classify the available data?

The MNIST laboratory shifted my attention toward a different but related question: how does information move from its original representation into a form that allows a model to learn useful features? Rather than looking only at the final classification result, I became more interested in the transformations that occur between the original data and that prediction. This gave me a stronger understanding of why data preparation, model architecture, and evaluation should be considered connected parts of one workflow.

The laboratory also clarified how the stages of a CNN work together. Data preparation establishes the input representation; convolutional layers examine local regions and learn features; pooling reduces spatial dimensions; and later layers use the learned representation to classify the digit. Seeing this sequence as one connected information-processing workflow made the architecture easier to understand. It also showed me that the model does not operate independently of the data. The structure created during preparation influences how the network can receive and transform the information.

My biggest takeaway is that successful deep learning begins before training. Preparing data correctly is part of building the model, not merely an administrative step. My earlier classification work taught me to compare models and interpret their performance, while the MNIST laboratory helped me examine more closely how information is represented and transformed before reaching the final classification. Together, these experiences gave me a broader understanding of machine learning workflows. I now see data preparation, representation, architecture, and evaluation as connected parts of the same process. In future projects, I want to examine datasets more deliberately before training so I can better explain not only whether a model succeeds, but how the preparation and representation of the data contribute to that result.